

**A REPORT
ON
AI-Based Smart Exam Proctoring and Malpractice
Detection**

Submitted by

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CERTIFICATE

This is to certify that the report of **CSE7000 – Internship** on “**AI GPU SUMMER INTERSHIP PROGRAM**” being submitted by “**SOHAN SP**” bearing roll number “**20241CSE0506**” in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in **Computer Science and Engineering** is a bonafide work carried out under our supervision.

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I hereby declare that the work, which is being presented in the report entitled “**AI GPU SUMMER INTERSHIP PROGRAM**” in partial fulfillment for the award of Degree of **Bachelor of Technology in Computer Science and Engineering**, is a record of my own investigations carried under the guidance of **Mr.Sakthi S**, Assistant Professor and **Ms.Shaik Salma Begum**, Assistant Professor, Presidency School of Computer Science and Engineering, Presidency University, Bengaluru.

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ABSTRACT

The AI-Based Smart Exam Proctoring and Malpractice Detection System is an intelligent online examination platform developed to provide a secure, automated, and efficient environment for conducting examinations. The system uses a large validated question bank from which questions are randomly selected for each examination attempt, reducing repetition and making the assessment more reliable. The examination is conducted within a fixed time limit, and student responses are automatically evaluated to generate scores, percentages, and final result.

The system incorporates Artificial Intelligence, Computer Vision, and Machine Learning techniques for continuous background monitoring during the examination. OpenCV and MediaPipe are used for face detection, head movement analysis, looking-away detection, face absence, and multiple-face detection. YOLOv8 is used to identify mobile phones and multiple persons when they are visible within the camera frame. The system also provides microphone-based audio activity monitoring as an additional supporting feature.

Browser activities such as tab switching, window focus changes, and fullscreen exit are monitored, with events recorded and webcam evidence captured for review. A risk-scoring mechanism classifies examination attempts based on detected activities. The Admin portal manages students, questions, examinations, results, proctoring events, evidence, and analytics. Overall, the system combines randomized question selection, evaluation, AI-based monitoring, evidence collection, and risk analysis to provide a secure examination environment while reducing the need for invigilation.

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CHAPTER 1

INTRODUCTION

1.1 CONTEXT AND BACKGROUND

The rapid growth of digital education has significantly increased the use of online examination platforms in universities, colleges, training institutions, and other educational environments. Online examinations provide flexibility, accessibility, automated evaluation, and efficient management of large numbers of students. However, conducting examinations remotely introduces several challenges related to examination integrity, student authentication, and detection of malpractice.

In a traditional examination environment, invigilators continuously observe students to identify suspicious activities such as unauthorized communication, use of mobile phones, looking away from the examination screen, multiple persons being present, or attempting to access unauthorized resources. In an online examination, continuous manual supervision becomes difficult because students may participate from different locations and monitoring a large number of candidates simultaneously requires significant human effort.

Recent developments in Artificial Intelligence (AI), Computer Vision, and Machine Learning have created opportunities to automate several aspects of online examination monitoring. Computer vision models can analyze webcam frames to detect faces, multiple persons, mobile phones, and changes in head orientation. Browser-based monitoring techniques can also identify activities such as tab switching, leaving fullscreen mode, and changes in browser focus.

Therefore, there is a growing need for intelligent online examination systems that combine examination management, automated evaluation, real-time monitoring, evidence collection, and risk analysis within a unified platform. Such a system can assist administrators and invigilators in identifying potentially suspicious examination behavior while reducing the need for continuous manual supervision.

1.2 PROBLEM STATEMENT

The increasing adoption of online examinations has created significant challenges in maintaining examination integrity. Students participating in remote examinations may engage in activities such as looking away from the screen, using mobile phones, having another person present, switching browser tabs, exiting fullscreen mode, or attempting to access unauthorized resources. Detecting such activities manually is difficult, particularly when a large number of students are taking an examination simultaneously.

Another major challenge is the absence of comprehensive evidence and risk analysis mechanisms in many conventional online examination platforms. Simply detecting an unusual activity may not provide sufficient information for an administrator to determine whether an examination attempt requires further investigation. There is therefore a need to record suspicious events, capture relevant evidence, and analyze the accumulated events to generate an understandable risk assessment.

In addition, examination management, question delivery, automatic evaluation, browser monitoring, webcam analysis, and administrative reporting are often implemented as separate components. This separation can increase system complexity and make it difficult for administrators to obtain a complete overview of an examination attempt.

Therefore, there is a need for an AI-based online examination platform capable of combining secure examination delivery, automated assessment, real-time webcam monitoring, browser activity monitoring, malpractice detection, evidence collection, and risk analysis within a single integrated system.

1.3 PROPOSED SOLUTION

To address these challenges, this project introduces an AI-Based Smart Exam Proctoring and Malpractice Detection System, an intelligent web-based examination platform designed to combine online examination functionality with automated AI-assisted proctoring.

The system provides a complete examination environment in which students can log in, access assigned examinations, answer questions, navigate between questions, and submit their responses. Questions can be presented through an interactive examination interface containing a question palette, answer selection controls, examination timer, and submission mechanism. The system can automatically evaluate objective-type answers and maintain examination results for administrative review.

The second major capability of the system is AI-based examination monitoring. The student's webcam is continuously analyzed using computer vision techniques to identify suspicious activities. The system detects face absence, multiple persons, mobile phones, and abnormal head movements. Head orientation is analyzed to identify repeated looking away from the examination screen.

Browser-level monitoring detects actions such as tab switching, focus changes, fullscreen exit, right-click attempts, and keyboard actions associated with developer tools. When activities are identified, events are recorded and webcam evidence can be captured for examination.

A risk-scoring mechanism combines detected events and their severity to classify examination attempts according to their overall level of suspicious activity. This enables administrators to prioritize examination attempts that require further investigation rather than manually reviewing every examination session.

1.4 AI-BASED WEBCAM MONITORING AND MALPRACTICE DETECTION

The webcam monitoring component forms an important part of the proposed proctoring system. OpenCV is used for real-time webcam capture and image processing, while computer vision and deep learning models are used to analyze the captured frames.

Face detection is performed to determine whether the authorized examination participant remains visible during the examination. If no face is detected for a specified period, the system records a face-absence event. The system also detects multiple faces within the camera frame, which may indicate the presence of another person during the examination.

A YOLO-based object detection model is used to identify objects such as mobile phones. When a potentially unauthorized device is detected, the corresponding event is recorded by the system. This allows administrators to identify examination sessions in which a mobile phone or similar object was visible.

Head movement analysis uses MediaPipe Face Mesh and OpenCV to estimate head orientation using pitch, yaw, and roll. Movements can identify looking-away behavior. The detected head movements are combined with other proctoring events to improve the overall assessment of suspicious examination activity. However, conditions may affect accuracy, so detected events are recorded for review rather than proof of malpractice.

1.5 BROWSER ACTIVITY MONITORING AND EVIDENCE COLLECTION

In addition to webcam-based monitoring, the system monitors selected browser activities that may indicate attempts to leave the examination environment. Browser events such as tab switching, browser window focus changes, and fullscreen exit are detected and recorded during the examination.

The examination interface encourages students to remain within the designated examination environment. Fullscreen functionality is used to provide a focused examination interface, while browser activity monitoring helps identify actions that may indicate attempts to access external resources.

When suspicious activity is detected, the system records the event type, timestamp, examination session, and student details. Webcam evidence can also be captured for additional context. Administrators can review the collected evidence through the evidence gallery, while examination replay and event timeline features help them understand suspicious activities during the examination session.

1.6 SYSTEM INTEGRATION AND MANAGEMENT

The different components of the proposed system are integrated through a Python Flask backend and a web-based frontend. Flask manages examination operations, student authentication, database communication, AI model inference, proctoring events, and evidence management. OpenCV, YOLOv8, and MediaPipe Face Mesh support webcam processing, object detection, facial landmark detection, and head movement analysis. The database manages students, examinations, questions, answers, results, and evidence. REST API endpoints facilitate communication between the frontend and backend, enabling smooth data exchange during examination sessions.

The system provides separate interfaces for students and administrators. Students can access examinations, answer questions, navigate through the question palette, and submit responses, while administrators can manage students, questions, examinations, results, events, and evidence. The administrative dashboard provides analytics, risk scores, event timelines, and evidence management. It also allows administrators to monitor examination activities and review suspicious events. Export functionality allows administrators to generate reports.

The system combines online examination management with AI-assisted monitoring in a unified platform. It includes automatic evaluation, webcam monitoring, malpractice detection, browser activity monitoring, evidence collection, and risk scoring. AI-based detections may produce false positives due to lighting, camera quality, student movement, or other factors. Therefore, suspicious events and risk scores are treated as indicators for administrative review rather than final proof of misconduct.

1.7 ORGANIZATION OF THE REPORT

This report presents the design, development, implementation, testing, and evaluation of the AI-Based Smart Exam Proctoring and Malpractice Detection System. It describes the background and challenges associated with online examinations, the proposed solution, system architecture, technologies used, implementation methodology, AI-based monitoring techniques, examination management features, and administrative functionality.

The report further discusses the results obtained from the implemented system, including examination processing, webcam-based detection, browser activity monitoring, evidence collection, risk analysis, and administrative management. Finally, the report presents the conclusion, limitations, future enhancements, and references used during the development of the project.

CHAPTER 2

OBJECTIVES

The main objective of the AI-Based Smart Exam Proctoring and Malpractice Detection System is to develop a secure and automated online examination platform that combines aptitude testing with artificial intelligence-based monitoring. The system is designed to conduct examinations in a controlled environment while reducing the need for continuous manual supervision. It provides student authentication, preexamination verification, randomized question selection, automatic evaluation, and continuous monitoring of suspicious activities during the examination.

2.1 EXAMINATION MANAGEMENT

To develop a secure online examination platform that allows students to log in and attend aptitude-based examinations through a simple and user-friendly interface with multiple-choice questions, question navigation, question palette, countdown timer, and examination submission. The system is designed to provide a structured examination environment where students can complete the test within the specified time. It also maintains the examination progress and responses throughout the attempt.

The system performs pre-examination verification by checking webcam and microphone availability and capturing a student selfie as baseline evidence before the examination begins. It also provides a structured examination environment with a countdown timer, examination progress tracking, and secure submission. These features help ensure that the examination begins under proper monitoring conditions and that the student's attempt is recorded securely for evaluation and further review.

2.2 AI-BASED PROCTORING AND RISK ASSESSMENT

To implement AI-based webcam monitoring using computer vision techniques to detect face absence, multiple faces, suspicious head movements, and unauthorized objects such as mobile phones during the examination. The camera monitoring operates in the background so that the student can focus on the examination questions. Detected suspicious conditions are processed and recorded for further examination. The monitoring system provides continuous observation of the examination environment without interrupting the student's examination experience.

To record suspicious activities and capture relevant evidence with details such as student identity, event type, timestamp, and severity, and use the detected events to calculate a risk score for each examination attempt. The collected evidence helps administrators understand the reason behind a particular suspicious event. The risk score provides an overall indication of the level of suspicious activity during the examination. This information supports administrators in reviewing examination attempts and identifying sessions that may require further investigation.

To provide automatic examination evaluation and an administrator module for examination management. The system evaluates the examination after submission or timer expiration by calculating attempted, correct, incorrect, and unanswered questions, along with the final score and percentage. The results are stored in the database. The administrator can review student attempts, results, proctoring events, risk scores, and captured evidence. This helps identify suspicious attempts and reduces the need for supervision.

CHAPTER 3

WORK PROGRESS

The AI GPU Summer Internship Program 2026 was completed over a period of two weeks, consisting of ten instructional days from 1 June 2026 to 12 June 2026. The internship provided practical and theoretical exposure to Artificial Intelligence, Machine Learning, Neural Networks, Deep Learning, Generative AI, Large Language Models, GPU computing, and software development. The program was conducted as an intensive training program with practical laboratory activities and project-oriented learning.

3.1 WEEK 1: AI FOUNDATIONS AND DEEP LEARNING ESSENTIALS

DAY 1 – ENVIRONMENT SETUP AND GPU COMPUTING

The first day focused on understanding the preparation of a suitable environment for AI and Machine Learning activities. The pod creation process was studied, including the allocation of CPU, memory, storage, and GPU resources. A GPU-enabled JupyterLab environment was launched and tested to verify notebook access, kernel readiness, and availability of required tools. The session highlighted the importance of proper environment configuration for stable and efficient AI development.

DAY 2 – ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING FUNDAMENTALS

The second day introduced the concepts of Artificial Intelligence and Machine Learning. The differences between AI, Machine Learning, Deep Learning, and automation were studied. Concepts such as data, features, models, training, testing, and prediction were discussed. Supervised and unsupervised learning techniques were explored through examples. The session helped establish a foundation for understanding intelligent systems.

DAY 3 – NEURAL NETWORKS

The third day focused on the fundamentals of neural networks and their role in pattern recognition. Artificial neurons, input layers, hidden layers, and output layers were studied in detail. The importance of weights, biases, and activation functions in neural network computation was explored. The session also covered how information moves through layers during prediction and how neural networks learn by adjusting their parameters. Examples were used to understand the structure and working process of neural networks. The session provided an understanding of how neural networks support AI applications.

DAY 4 – DEEP LEARNING

The fourth day focused on Deep Learning and multi-layer neural network architectures. The relationship between traditional Machine Learning and Deep Learning was studied, along with the role of hidden layers in extracting increasingly complex features. Applications of Deep Learning in image recognition, speech processing, and language-related tasks were discussed. The importance of large datasets and computational resources for training deep learning models was also examined.

DAY 5 – ANTIGRAVITY AND FULL-STACK DEVELOPMENT

The fifth day introduced the Antigravity tool and provided an overview of software development and integration concepts. Front-end, back-end, and database components were studied along with their roles in developing complete software applications. The interaction between user interfaces, server-side applications, and databases was discussed. The day also included revision of the concepts covered during the first week and preparation for the advanced AI topics in the second week.

3.2 WEEK 2: ADVANCED AI AND GPU OPTIMIZATION

DAY 6 – TRANSFORMER ARCHITECTURE

The sixth day focused on Transformer architecture and attention mechanisms used in modern Artificial Intelligence systems. Encoder and decoder structures, self-attention, and multi-head attention were studied. Transformers were compared with traditional recurrent neural network architectures to understand their advantages in processing sequential information. The session also explored the use of Transformers in Natural Language Processing and contextual information processing.

DAY 7 – LARGE LANGUAGE MODELS AND FINE-TUNING

The seventh day introduced Large Language Models (LLMs) and techniques for adapting them to specialized applications. LoRA and PEFT fine-tuning methods were studied along with transfer learning and model customization. The session provided an understanding of how efficient fine-tuning techniques can reduce computational requirements while allowing models to be adapted for specific tasks. Model deployment and optimization concepts were also discussed. Practical examples demonstrated how customized language models can be applied to real-world applications.

DAY 8 – GENERATIVE AI AND IMAGE GENERATION

The eighth day focused on Generative AI and its applications. Stable Diffusion XL and text-to-image generation workflows were explored. Prompt engineering techniques were studied to understand how prompts influence the quality and relevance of generated outputs. Prompt styles and generated results were compared to understand the importance of prompt design in Generative AI applications. The session also highlighted the role of Generative AI in creating creative digital content.

DAY 9 – GPU OPTIMIZATION

The ninth day focused on GPU optimization and improving the efficiency of AI model execution. Concepts such as mixed precision training, memory optimization, FlashAttention, and tensor parallelism were studied. GPU resource utilization and performance improvement techniques were also discussed. The session provided an understanding of how optimization methods can improve training speed, memory efficiency, scalability, and overall AI system performance.

DAY 10 – PROJECT PRESENTATION AND PROGRAM COMPLETION

The final day was dedicated to presenting the completed project and reviewing the knowledge gained during the internship. The project implementation process was explained, followed by technical presentations and peer review discussions. Feedback from mentors and trainers was received to identify areas for improvement. The session also emphasized the importance of communication, teamwork, technical explanation, and problem-solving skills. The internship concluded with the closing activities and completion of the training program.

3.3 OVERALL WORK PROGRESS

The two-week internship provided a progressive learning experience, beginning with GPU environment setup, Artificial Intelligence, and Machine Learning fundamentals and advancing toward Neural Networks, Deep Learning, Transformers, Large Language Models, Generative AI, and GPU optimization. The practical sessions helped develop an understanding of AI development environments and modern AI technologies. The internship also provided exposure to software architecture and application integration through the study of the Antigravity tool and full-stack development concepts.

CHAPTER 4

RESULTS

4.1 ONLINE TEST INTERFACE

Figure 4.1 shows the main Online Test interface of the proposed system. It displays the current question, multiple-choice options, countdown timer, and question palette for navigation. The interface also shows the active proctoring, face detection, and microphone status. The Previous, Save & Next, and Submit Exam buttons allow the student to navigate and complete the test. The simple layout keeps the student focused on the test while monitoring continues in the background. The question palette helps the student quickly move between different questions during the examination. It also provides a clear view of the examination progress and supports smooth test navigation. The interface ensures that the examination can be completed in an organized and user-friendly manner.

4.2 ADMIN DASHBOARD

Figure 4.2 shows the Admin Dashboard of the AI-Based Smart Exam Proctoring and Malpractice Detection System. It provides administrators with an overview of students, completed examinations, average scores, and high-risk examination attempts. The dashboard also displays recent exam attempts along with student details, examination date, score, percentage, risk score, and risk status. The Low Risk, Medium Risk, and High Risk classifications help administrators quickly identify suspicious examination attempts. The **View Details** option provides access to additional information about each attempt and its recorded proctoring events. Overall, the dashboard enables efficient monitoring of examination activities, student performance, and potential malpractice.

4.3 PROCTORING ATTEMPT DETAILS

Figure 4.3 shows the Attempt Details page of the system, where the administrator can review a completed examination. It displays the student's examination information, final score, percentage, and overall risk score with its risk level. The Proctoring Events Log records detected activities such as no face, fullscreen exit, and tab switching along with their time and severity. Evidence images can also be viewed for relevant events, helping the administrator review suspicious activities after the test. The page provides a centralized view of the student's examination performance and monitoring information. It helps the administrator assess the examination attempt and take appropriate action when required. This detailed information supports effective identification and verification of potential malpractice during the examination.

4.4 EXAM RESULT PAGE

Figure 4.4 shows the Exam Result page of the proposed system, where the student can view the outcome of the completed examination. It displays the total number of questions, attempted questions, correct answers, wrong answers, and unanswered questions. The page also presents the final score and percentage achieved by the student. The Proctoring Risk Score indicates the level of suspicious activity detected during the examination. In this example, the attempt is classified as High Risk based on the recorded proctoring activities. The result page provides a clear summary of both examination performance and monitoring status. It helps the student understand their test performance and the associated proctoring assessment. The Logout button allows the student to safely exit the examination system after viewing the result.

ONLINE APTITUDE TEST

Student: student001

Time Remaining: 59:11

Question 45 of 50

A can do a piece of work in 20 days and B can do it in 20 days. How long will they take if they work together?

☐ 10.00 days
 ☐ 20.00 days
 ☐ 10.00 days
 ☐ 12.00 days

Previous

Save & Next

Proctoring: Active

Face: / Detected

Microphone: Active

Question Palette

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	25
26	27	28	29	30
31	32	33	34	35
36	37	38	39	40
41	42	43	44	45
46	47	48	49	50

Submit Exam

Figure 4.1: Online Test Interface

Admin Dashboard

Logout

1
Total Students

6
Exams Completed

0.33%
Average Score

4
High Risk Attempts

Recent Exam Attempts

Student	Date	Score	Percentage	Risk Score	Status	Action
Demo Student (student001)	2026-08-10 18:33:17	-	-	5	Low Risk	View Details
Demo Student (student001)	2026-08-10 10:43:54	0	0.00%	8460	High Risk	View Details
Demo Student (student001)	2026-08-09 23:30:09	0	0.00%	50	Medium Risk	View Details
Demo Student (student001)	2026-08-09 20:27:41	-	-	50	Medium Risk	View Details
Demo Student (student001)	2026-08-09 19:43:19	0	0.00%	115	High Risk	View Details
Demo Student (student001)	2026-08-09 19:39:33	-	-	50	Medium Risk	View Details
Demo Student (student001)	2026-08-09 19:28:54	0	0.00%	155	High Risk	View Details
Demo Student (student001)	2026-08-09 19:16:24	0	0.00%	10	Low Risk	View Details
Demo Student (student001)	2026-08-09 19:11:52	1	2.00%	365	High Risk	View Details

Figure 4.2: Admin Dashboard

15

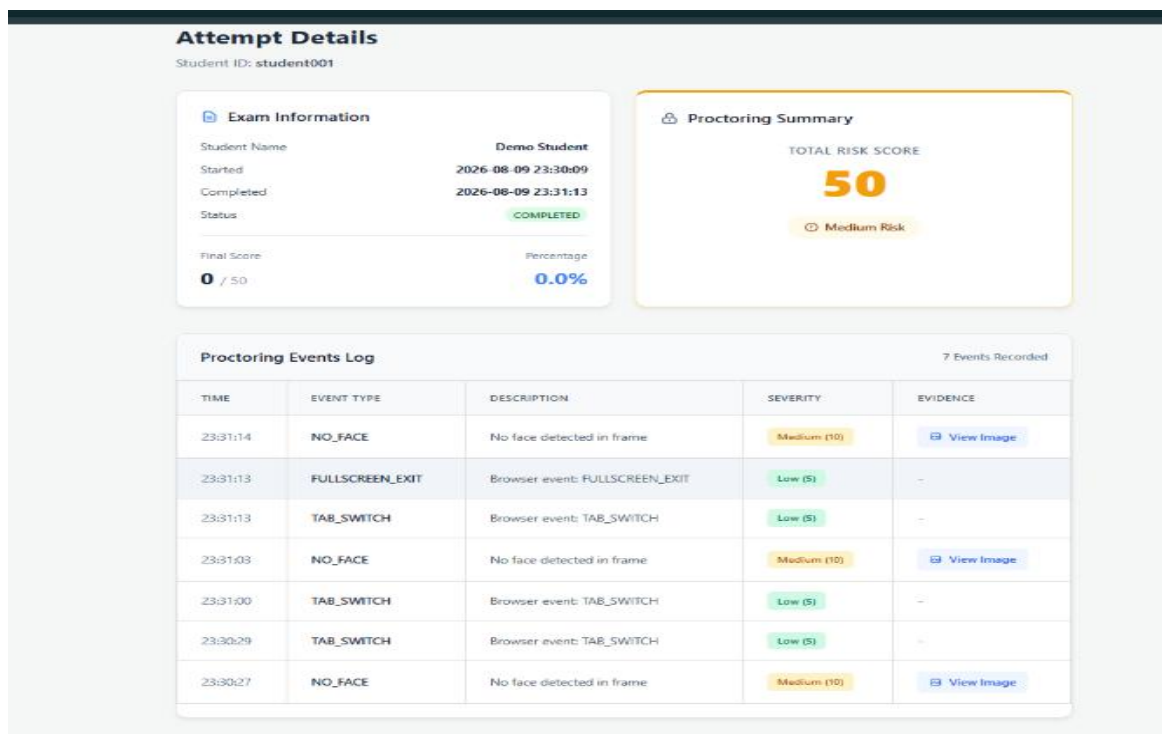


Figure 4.3: Proctoring Attempt Details

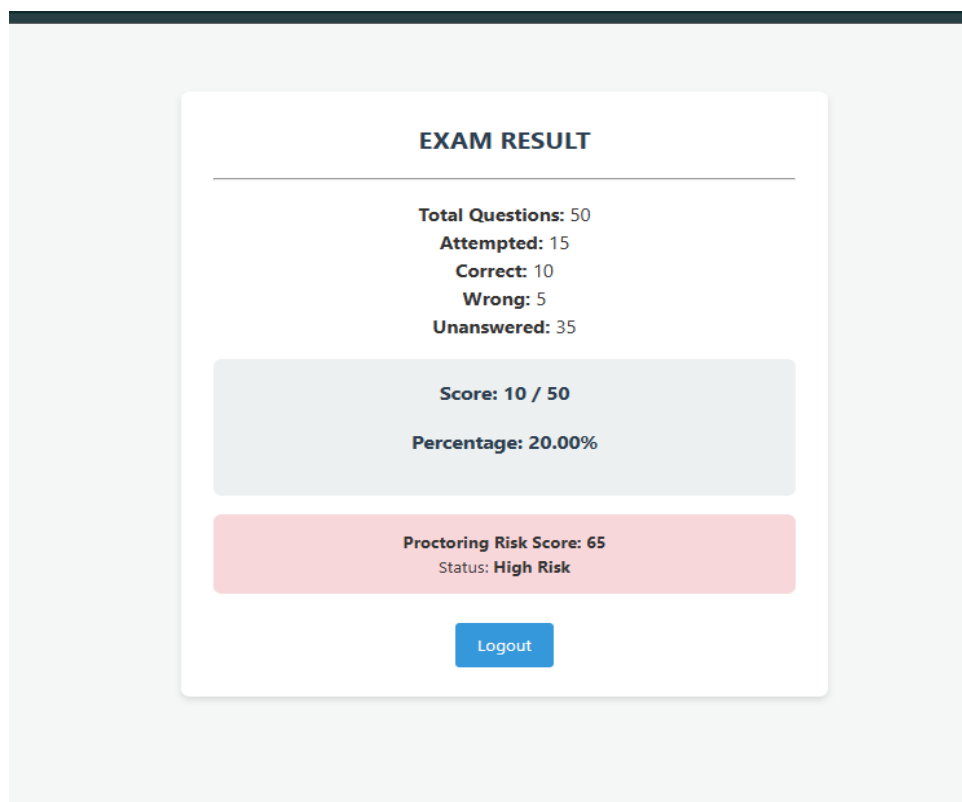


Figure 4.4: Exam Result Page

CHAPTER – 5

CONCLUSION

5.1 CONCLUSION

The AI-Based Smart Exam Proctoring and Malpractice Detection System demonstrates the feasibility of integrating artificial intelligence and computer vision techniques into an automated online examination platform. By combining examination management, automatic evaluation, webcam monitoring, suspicious activity detection, evidence collection, and risk assessment, the system provides a unified platform for conducting secure online examinations and reviewing examination activities.

The system continuously monitors examination activities and detects conditions such as face absence, multiple faces, suspicious head movements, mobile phone presence, tab switching, and fullscreen exit. Detected events are recorded with relevant details and supporting evidence to generate a risk score for each examination attempt. The Admin portal provides centralized access to examination results, proctoring events, and evidence. Overall, the system reduces the need for continuous manual invigilation and provides a more secure and reliable examination environment.

5.2 FUTURE SCOPE

The current system provides a foundation for future improvements. The proctoring module can be enhanced with advanced deep-learning models for better detection accuracy and reduced false alerts. Facial recognition, audio monitoring, and improved behavioral analysis can be added for stronger identity verification and malpractice detection. Further improvements may include real-time administrator alerts, examination replay, detailed analytics, and cloud-based deployment to improve the scalability, reliability, and effectiveness of the proposed system.

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